**1** Write $\frac{2}{5}$ as a decimal.

0.4

**(Total for Question 1 is 1 mark)**

---

**2** Write 0.03 as a fraction.

$\frac{3}{100}$

**(Total for Question 2 is 1 mark)**

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**3** 5 calculators cost £31.25 $5 \times 6 = 30$
3 pens cost £2.91 $3 \times 10 = 30$

1 mark for correct method

Work out the cost of 30 calculators and 30 pens.

$31.25 \times 6 + 2.91 \times 10$

$= £216.60$

£ 216.60

**(Total for Question 3 is 2 marks)**

---

**4** A square has a perimeter of 32 cm.

Find the area of the square.

$\frac{32}{4} = 8$ each side is 8cm

1 mark for 8 (cm)

$8 \times 8 = 64 \text{ cm}^2$

64 $\text{cm}^2$

**(Total for Question 4 is 2 marks)**

**5** Here are six cards. Each card has a number on it.

| | | | | | |
|---|---|---|---|---|---|
| 15 | 14 | 9 | 12 | 16 | 21 |

(a) Work out the range of the numbers on the cards.

$21 - 9$

12

(1)

(b) Work out the mean of the numbers on the cards.

$$\frac{15 + 14 + 9 + 12 + 16 + 21}{6} = 14.5$$

1 mark for correct method

14.5

(2)

**(Total for Question 5 is 3 marks)**

---

**6** Work out $\dfrac{3.15 + 2.8^2}{2.06}$

1 mark for 10.99

Write down all the figures on your calculator display.

5.334951456

**(Total for Question 6 is 2 marks)**

---

**7** $L = 9m + 2n$

Work out the value of $L$ when $m = -3$ and $n = 4$

$9(-3) + 2(4) = -19$

1 mark for correct substitution

$-19$

**(Total for Question 7 is 2 marks)**

**8** (a) Solve $3x = 36$

$x =$ 12

(1)

(b) Solve $\frac{24}{y} = 6$

$y =$ 4

(1)

(c) Solve $a - 8 = 17$

$a =$ 25

(1)

**(Total for Question 8 is 3 marks)**

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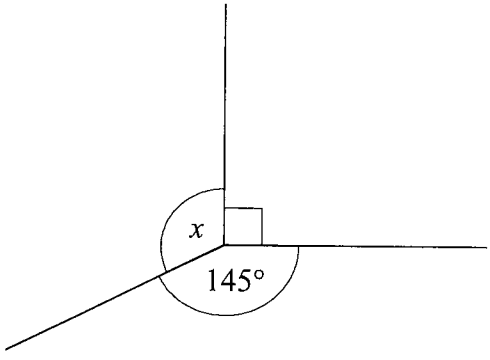
**9** Work out the size of the angle marked $x$.



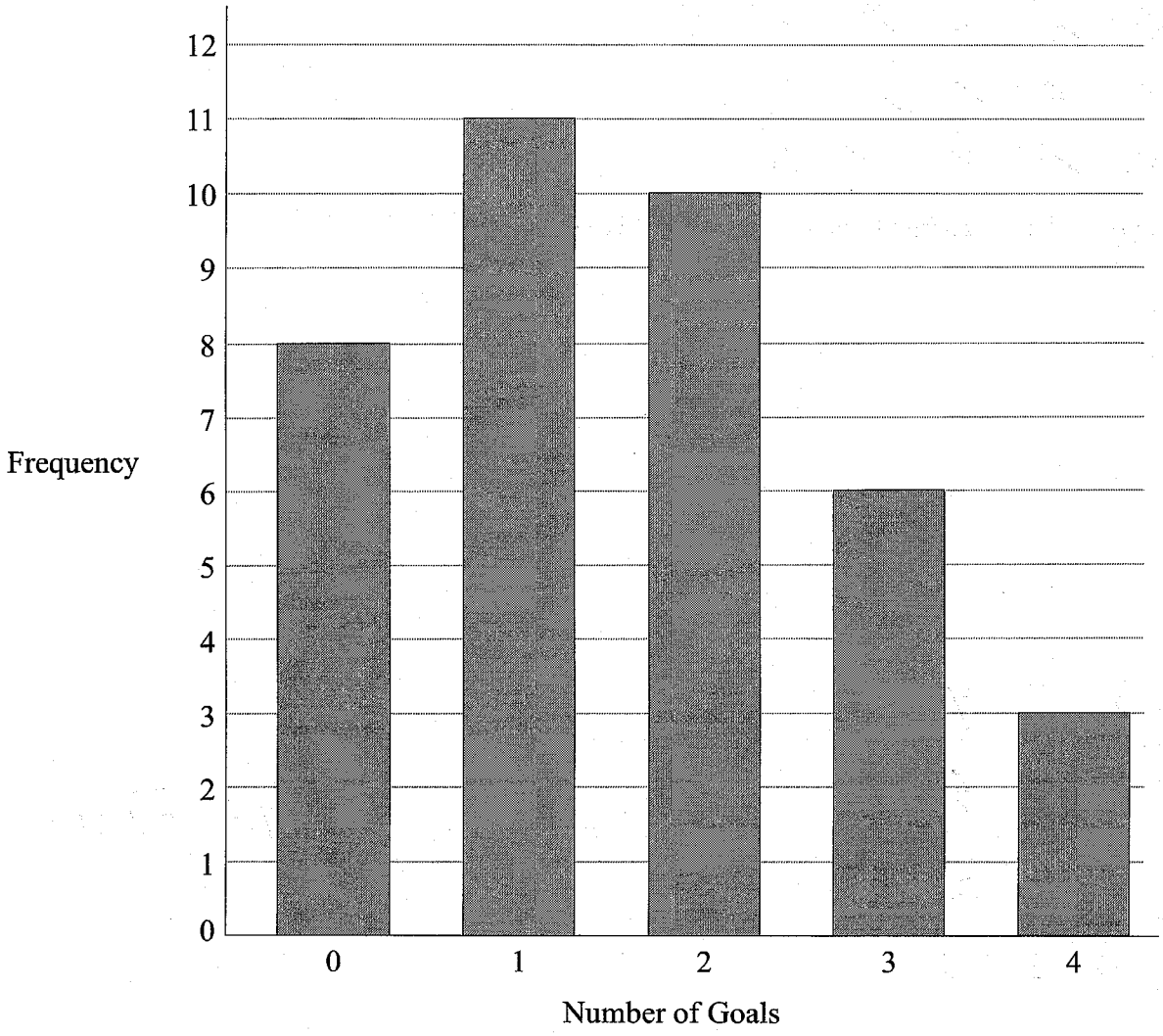

360 − 90 − 145

1 mark for correct method

125 °

**(Total for Question 9 is 2 marks)**

**10** Here is a bar chart showing the number of goals scored in a game by a football team in a season.




(a) What is the modal number of goals scored.

1

(1)

(b) Work out the total number of goals were scored in total by the team.

1 mark for (8), 11, 10, 6 and 3

$11 \times 1 = 11$

$10 \times 2 = 20$

$6 \times 3 = 18$

$3 \times 4 = 12$

$11 + 20 + 18 + 12 = 61$

61

(3)

1 mark for correct method

**(Total for Question 10 is 4 marks)**

**11** Here are the heights, in cm, of 15 plants.

| | | | | |
|---|---|---|---|---|
| 50 | 39 | 52 | 53 | 25 |
| 32 | 29 | 45 | 35 | 33 |
| 29 | 27 | 46 | 44 | 41 |

Draw an ordered stem and leaf diagram to show this information.

1 mark for just one mistake

| | |
|---|---|
| 2 | 5 7 9 9 |
| 3 | 2 3 5 9 |
| 4 | 1 4 5 6 |
| 5 | 0 2 3 |

Key: 2/5 = 25cm

1 mark for correct key

**(Total for Question 11 is 3 marks)**